

Bofah Dennis

(416) 836-0247 | bofah.d@northeastern.edu | Toronto, ON

QUALIFICATIONS

I am a Master of Science in Informatics, Data Analytics student at Northeastern University with a background in Electrical and Electronics Engineering. I am skilled in data engineering, analytics, and cloud technologies, including Python, R, SQL, Apache Spark, and AWS. Experienced in optimizing ETL pipelines, automating workflows, and enhancing system performance and security. Strong collaborator in cross-functional teams, deliver data-driven insights and process improvements. Demonstrated ability in technical reporting, mentoring, and implementing scalable data solutions.

EDUCATION

Northeastern University

Toronto, ON

Master of Science in Informatics, Data Analytics

Expected Graduation Date: Jun 2026

Academics: Cumulative GPA of **3.8/4.0**

Relevant Coursework: Developing expertise in Data Collection, Data Cleaning, Data Exploration, and Transformation.

University of Mines and Technology

Ghana

Bachelor of Science in Electrical and Electronics Engineering

July 2016 – August 2019

Academics: Cumulative GPA of **3.9/4.0**

Relevant Course Work: Data-driven decision-making, system optimization, and analytics.

Professional Experience

VRE/NEDCO

Ghana

Electrical Systems Technician

September 2022 – August 2024

- Optimized SQL queries, improving data retrieval speed by 30% and reducing report generation time by 20%.
- Automated data cleaning processes using Python, reducing manual effort by 25%.
- Collaborated with cross-functional teams to design and implement a cloud-based data pipeline using AWS (S3, Lambda).
- Gained hands-on experience with big data processing tools like Apache Spark and Airflow.

Technical Project

HVALIDATE Healthcare Analytics Dashboard (Capstone Project)

Northeastern University – Master of Professional Studies in Informatics.

January 2026 – March 2026

- * Designed and implemented a healthcare analytics dashboard to support clinical decision-making by visualizing patient severity stages, validation rates, and data inconsistencies.
- * Developed data processing workflows using Python and SQL to clean, validate, and transform healthcare datasets, ensuring high data quality and reliable analytics outputs.
- * Integrated secure cloud-based data architecture using Google Cloud Platform (GCP) services to enable scalable storage, analytics, and dashboard deployment.
- * Implemented data governance and security controls, including HIPAA-aligned privacy measures and role-based access management (IAM), to protect sensitive healthcare information.
- * Designed validation metrics to monitor data consistency, validation success rates, and cohort-based health outcomes, enabling healthcare administrators to identify data quality issues and improve reporting accuracy.

Ghana Highway Authority (National Service)

Electronic Toll Collection System Project.

November 2019 – August 2021

- Designed and implemented an optimized data flow model for efficient toll collection, reducing processing time by 15%.
- Developed ETL workflows using Python and SQL to clean and process toll transaction data, ensuring 99% data accuracy.
- Integrated cloud-based analytics (AWS Redshift) to enhance system performance and reporting.
- Implemented security measures that reduced fraudulent transactions by 25%.

COMMUNITY INVOLVEMENT

STEM Outreach Program Ghana

Volunteer Mentor

March 2022 – August 2022.

- Provided mentorship to high school students interested in **STEM and data analytics**.
- Conducted **workshops on Python and SQL**, helping students develop fundamental programming skills.
- Assisted in organizing community hackathons to promote **data-driven problem-solving**.
- Participated in **data engineering workshops** on cloud-based analytics and scalable data pipelines.

TECHNICAL SKILLS & INTERESTS

Technical Skills:

Programming: Python, SQL, Apache Spark, Airflow, Cloud Platforms: AWS, GCP, Azure, Data Engineering: ETL Pipelines, Data Warehousing, NoSQL Databases, Tools & Technologies: MySQL, Microsoft Office Suite (Excel, etc.), Tableau.

Interests: Competitive Cycling, Athlete & Team Coaching, Meditation, Fitness, Reading, listening to music, Strategy, and Operations